Arithmetic and Geometric Series – Worksheet #2

MCR3U

SOLUTIONS

1) The first and last terms in each arithmetic series are given. Determine the sum of the series.

a) $a = \frac{1}{2}$ and $t_8 = 4$

$S_n = \frac{n}{2}(a + t_n)$

$S_8 = \frac{8}{2}(\frac{1}{2} + 4)$

$S_8 = 18$

b) $a = 11$ and $t_{20} = 101$

$S_{20} = \frac{20}{2}(11 + 101)$

$S_{20} = 1120$

2) Determine the sum of the arithmetic series $-1+2+5+\ldots+164$. (13, t_{56})

$t_n = a + (n-1)d$

$164 = -1 + (n-1)(3)$

$165 = (n-1)(3)$

$55 = n-1$

$n = 56$

$S_{56} = \frac{56}{2}(-1 + 164)$

$= 4564$

3) The 15th term in an arithmetic sequence is 43 and the sum of the first 15 terms of the series is 120. Determine the first three terms of the sequence.

$t_n = a + (n-1)d$

$43 = a + (15-1)d$

① $43 = a + 14d$

$S_{15} = \frac{15}{2}[2a + (15-1)d]$

$120 = 7.5(2a + 14d)$

② $120 = 15a + 105d$

15×① $645 = 15a + 210d$

② $120 = 15a + 105d$

$525 = 105d$

$d = 5$

$43 = a + 14d$

$43 = a + 14(5)$

$43 - 70 = a$

$a = -27$

$t_n = -27 + (n-1)(5)$

∴ first three terms are: $-27, -22, -17$

4) A toy car is rolling down an inclined track and picking up speed as it goes. The car travels 4 cm in the first second, 8 cm in the second second, 12 cm in the next second, and so on. Determine the total distance travelled by the car in 30 seconds.

$a = 4$

$d = 4$

$S_{30} = \frac{30}{2}[2(4) + (30-1)(4)]$

$= 15[8 + 29(4)]$

$= 1860$

∴ The car travels 1860 cm in 30 seconds.

5) For each geometric series, determine the values of a and r. Then, determine the indicated sum.

a) S_8 for $2 + 6 + 18 + \cdots$

$a = 2$, $r = 3$

$$S_8 = \frac{2(3^8 - 1)}{3 - 1}$$

$$= \frac{2(6560)}{2}$$

$$= 6560$$

b) S_{10} for $24 - 12 + 6 - \cdots$

$a = 24$, $r = -\frac{1}{2}$

$$S_{10} = \frac{24\left[\left(-\frac{1}{2}\right)^{10} - 1\right]}{-\frac{1}{2} - 1}$$

$$= \frac{24\left(\frac{1}{1024} - \frac{1024}{1024}\right)}{-\frac{3}{2}}$$

$$= \frac{-24552}{1024} \cdot -\frac{2}{3}$$

$$= \frac{49104}{3072}$$

$$= \frac{1023}{64}$$

6) Determine the sum of the geometric series $\frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \frac{8}{81} + \cdots + \frac{128}{6561}$

$a = \frac{1}{3}$, $r = \frac{2}{3}$

$t_n = ar^{n-1}$

$$\frac{128}{6561} = \frac{1}{3}\left(\frac{2}{3}\right)^{n-1}$$

$$\frac{384}{6561} = \left(\frac{2}{3}\right)^{n-1}$$

$$\frac{128}{2187} = \left(\frac{2}{3}\right)^{n-1}$$

$$\left(\frac{2}{3}\right)^7 = \left(\frac{2}{3}\right)^{n-1}$$

$$7 = n - 1$$

$$n = 8$$

$$S_8 = \frac{\left(\frac{1}{3}\right)\left[\left(\frac{2}{3}\right)^8 - 1\right]}{\frac{2}{3} - 1}$$

$$= \frac{\frac{1}{3}\left(\frac{256}{6561} - \frac{6561}{6561}\right)}{-\frac{1}{3}}$$

$$= -1\left(\frac{-6305}{6561}\right)$$

$$= \frac{6305}{6561}$$

7) Determine the sum of the geometric series $5 - 15 + 45 - \cdots + 3645$

$a = 5$, $r = -3$

$t_n = ar^{n-1}$

$$3645 = 5(-3)^{n-1}$$

$$729 = (-3)^{n-1}$$

$$(-3)^6 = (-3)^{n-1}$$

$$6 = n - 1$$

$$n = 7$$

$$S_7 = \frac{5\left[(-3)^7 - 1\right]}{-3 - 1}$$

$$= \frac{5(-2188)}{-4}$$

$$= 2735$$

8) The sum of $4 + 12 + 36 + 108 + \cdots + t_n$ is 4372. How many terms are in this series?

$a = 4$, $r = 3$ (×3)

$$4372 = \frac{4(3^n - 1)}{3 - 1}$$

$$8744 = 4(3^n - 1)$$

$$2186 = 3^n - 1$$

$$2187 = 3^n$$

$$3^7 = 3^n$$

$$n = 7$$

9) The third term of a geometric series is 24 and the fourth term is 36. Determine the sum of the first 10 terms.

$t_3 = ar^{3-1}$

$24 = ar^2$

$a = \frac{24}{r^2}$

$t_4 = ar^{4-1}$

$36 = ar^3$

$36 = \left(\frac{24}{r^2}\right)r^3$

$36 = 24r$

$r = \frac{36}{24}$

$r = \frac{3}{2}$

$a = \frac{24}{\left(\frac{3}{2}\right)^2}$

$= \frac{24}{\left(\frac{9}{4}\right)}$

$= 24 \cdot \frac{4}{9}$

$= \frac{32}{3}$

$$S_{10} = \frac{\frac{32}{3}\left[\left(\frac{3}{2}\right)^{10} - 1\right]}{\frac{3}{2} - 1}$$

$$= \frac{\frac{32}{3}\left[\frac{59049}{1024} - \frac{1024}{1024}\right]}{\frac{1}{2}}$$

$$= \frac{64}{3}\left(\frac{58025}{1024}\right)$$

$$= \frac{58025}{48}$$

Answers

1) a) 18 **b)** 1120

2) 4564

3) -27, -22, -17

4) 1860 cm

5) a) $S_8 = 6560$ **b)** $S_{10} = \frac{1023}{64}$

6) $S_8 = \frac{6305}{6561}$

7) $S_7 = 2735$

) 7 terms

9) $S_{10} = \frac{58025}{48}$